

Cross-Sectional Associations Between Food Addiction Symptoms and Cardiometabolic Biomarkers in Young Adults

Ariel Ooms, Vivienne M. Hazzard

Michigan State University

Background

- Cardiometabolic consequences of food addiction have not been well studied
- Possible links between food addiction and cardiometabolic biomarkers have been identified, but results have been inconclusive
- Advancing our understanding of how food addiction affects the human body could impact clinical practice for patients with food addiction symptoms

Study Objective

To identify key associations between food addiction symptoms and cardiometabolic biomarkers, as well as the extent to which those associations exist independent of Body Mass Index (BMI)

Methods

Participants

- The analytic sample is comprised of 372 young adults in the NEXT Plus subsample of the NEXT Generation Health Study with data available on basic demographic characteristics, food addiction symptoms, cardiometabolic biomarkers in Wave 7 of the longitudinal cohort study (collected in 2015-2016)



Measures

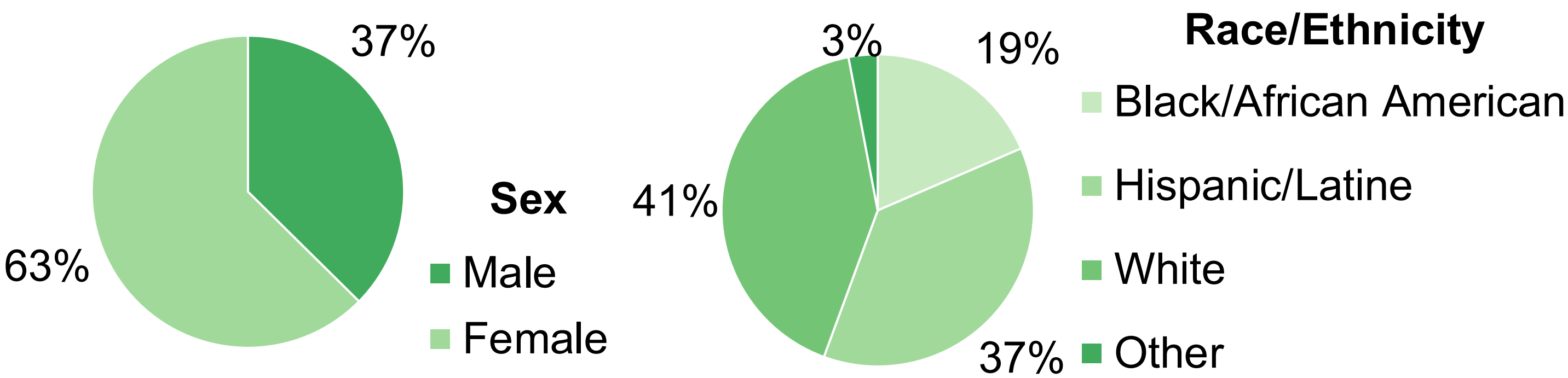
- Independent variable: Food addiction symptoms were assessed with the modified Yale Food Addiction Scale (mYFAS), with a food addiction symptom count variable representing the sum of dichotomous responses to the 7 mYFAS symptom variables (possible range: 0-7; sample mean = 0.9 ± 1.4)
- Dependent variables: Assessed via blood samples, dependent variables include high-sensitivity C-reactive protein (hsCRP), total cholesterol to high-density lipoprotein (HDL) ratio, triglycerides, fasting blood glucose, hemoglobin A1c (HbA1c), and alanine aminotransferase (ALT)

Statistical Analysis

- Separate linear regression models regressed each biomarker (transformed as necessary to meet normality assumptions) on food addiction symptom score, adjusting for age, sex, and race/ethnicity
- A second set of regression models additionally adjusted for BMI

Baseline Sample Characteristics

The baseline sample was, on average, 22.6 ± 0.6 years of age, was comprised of primarily female participants, and was racially/ethnically diverse



Results

	Unadjusted for BMI	Adjusted for BMI
	β (95% CI)	
hsCRP	0.12 (0.02, 0.23)*	0.01 (-0.08, 0.10)
Total Cholesterol to HDL Ratio	0.11 (0.01, 0.21)*	0.02 (-0.08, 0.12)
Triglycerides	0.06 (-0.05, 0.16)	0.00 (-0.11, 0.10)
Fasting Blood Glucose	0.03 (-0.08, 0.13)	0.00 (-0.11, 0.11)
HbA1c	0.04 (-0.07, 0.14)	0.00 (-0.11, 0.12)
ALT	0.01 (-0.08, 0.11)	-0.04 (-0.14, 0.05)

Note. All models were adjusted for age, sex, and race/ethnicity.

* $p < .05$.

Conclusions

- There are slight associations between food addiction and high-sensitivity C-reactive protein and total cholesterol to high-density lipoprotein (HDL) ratio
- Results indicate that these associations can be explained by BMI

Acknowledgments

This research was supported by the NICHD intramural research program, as well as by NHLBI, NIAAA, and the Maternal and Child Health Bureau of HRSA, with supplemental support from NIDA.